



DHARMSINH DESAI UNIVERSITY, NADIAD  
FACULTY OF TECHNOLOGY  
FIRST SESSIONAL

SUBJECT: (CE 313) Database Management Systems

Examination : B.Tech Semester 3

Date : 02, August 2022

Time : 1 hr 15 mins

Seat No. :

Day : Tuesday

Max. Marks : 36

**INSTRUCTIONS:**

1. Figures to the right indicate maximum marks for that question.
2. The symbols used carry their usual meanings.
3. Assume suitable data, if required & mention them clearly.
4. Draw neat sketches wherever necessary.

- Q.1 Do as directed. [12]
- CO2 (a) Examine: "Weak entity set needs to have total participation." Mention an example for the same. [2]
- CO1 (b) Explain with example: "For all relations, domains of all attributes need to be atomic". [2]
- CO1 (c) Describe the difference between super key and candidate key with example. [2]
- CO3 (d) Write a SQL statement to create a relation alpha\_students from the relation students without copying any values from the relation students. [2]
- (e) Consider the schema below. [2]
- Employee(empId, name, departmentNo, salary)
- CO3 Write SQL Queries for the following:
- i. Find the details of the employee(s) with the 3rd highest salary.
  - ii. Find the average salary in each department.
- CO1 (f) Differentiate DELETE and TRUNCATE commands. [2]
- Q.2 Attempt Any TWO from the following questions. [12]
- (a) Consider the given description and draw an E-R diagram: [6]
- There is a car insurance company whose customers own one or more cars each. Each car has associated with it zero to any number of recorded accidents. Each insurance policy covers one or more cars, and has one or more premium payments associated with it. Each payment is for a particular period of time, and has an associated due date as well as the date when the payment was received.
- CO2 (b) Consider the following relational database schema consisting of the four relation schemas: [6]
- passenger ( pid, pname, pgender, pcity)  
agency ( aid, aname, acity)  
flight ( fid, fdate, time, src, dest)  
booking ( pid, aid, fid, fdate)
- CO2 Answer the following questions using relational algebra:
1. Get the Flight Dates of all flights from Chennai to New Delhi.
  2. Find the passenger names for passengers who have bookings on at least one flight.
  3. Get the details of flights that are scheduled on both dates 02/08/2022 and 03/08/2022 at 16:00 hours.
- (c) Describe the following ways of reduction from E-R model to relation schema with appropriate examples: [6]
- CO2 Strong entity set which contains composite and derived attributes  
Strong entity set which contains multi valued attributes  
Weak entity set which contains simple attributes



Q.3 Attempt the following questions [12]  
CO1 (a) Write a SQL command for creating following relation 'students'. [02]

|               |             |                                                                            |
|---------------|-------------|----------------------------------------------------------------------------|
| s_id          | varchar(10) | must start with 'D'                                                        |
| s_name        | varchar(30) | not null                                                                   |
| s_branch      | varchar(5)  | value can be one of the following: CE, EC, IT, CH, MH, CL                  |
| s_nationality | varchar(25) | default: indian                                                            |
| s_dob         | date        | not null                                                                   |
| s_doa         | date        | not null, date of admission (doa) must be greater than date of birth (dob) |

CO3 (b) Write a SQL command for making s\_id primary key of student relation. [01]  
CO1 (c) Write a SQL command for creating following relation 'grades'. [02]

|        |             |                                                     |
|--------|-------------|-----------------------------------------------------|
| stu_id | varchar(10) | foreign key to s_id of students                     |
| marks  | number(5,2) | cannot be negative                                  |
| grade  | char(1)     | value can be one of the following: a, b, c, d, e, f |

What is the reason for following error?

"column type incompatible with referenced column type"

CO3 (d) Can following statement execute successfully for the students relations defined in [02]  
(a)? Justify your answer.

insert into students (S\_ID, S\_NAME, S\_BRANCH, S\_NATIONALITY, S\_DOB, S\_DOA) values ('D220223', 'abhay sharma', 'ce', ' ', '26-JAN-2002', '07-JUL-2001');

CO3 (e) Data for the relation students are as in Figure 1 [01]  
Write a SQL statement to display the number of students for each nationality who has a birthday between Feb 01 2002 and Oct 30 2002.

CO3 (f) Data for the relation students are as in Figure 1 and for the relation grade are as in [02]  
Figure 2.

Write a SQL statement to update the marks of student with stu\_id D220223 with the average marks of students whose id ends with 23 and are not in either EC or CH branch.

CO3 (g) Write a SQL statement to create a view student\_grade\_ch that contains all the [02]  
information (based on relations students and grades) of students whose s\_branch is CH.

| S_ID    | S_NAME            | S_BRANCH | S_NATIONALITY | S_DOB     | S_DOA     |
|---------|-------------------|----------|---------------|-----------|-----------|
| D220223 | abhay sharma      | CE       | india         | 26-JAN-02 | 07-JUL-22 |
| D220323 | nivedita patni    | EC       | canada        | 02-MAR-02 | 07-JUL-22 |
| D220329 | anju george       | EC       | india         | 12-MAR-02 | 07-JUL-22 |
| D220245 | shivani abhyankar | CE       | india         | 10-DEC-02 | 08-JUL-22 |
| D220445 | Farooq Shah       | CH       | india         | 20-OCT-02 | 08-JUL-22 |
| D220498 | avina dosa        | CH       | india         | 25-DEC-02 | 08-JUL-22 |

Figure 1

| STU_ID  | MARKS | GRADE |
|---------|-------|-------|
| D220223 | 85    | a     |
| D220223 | 75    | b     |
| D220223 | 79    | b     |
| D220323 | 65    | c     |
| D220445 | 94    | a     |
| D220245 | 91    | a     |
| D220329 | 56    | d     |
| D220329 | 71    | b     |

Figure 2

OR

Q.3 Attempt the following questions.

CO3 (a) Consider the relational database with the following four schemas and their [12]  
respective instances. [02]

Student(sNo, sName, dNo) Dept(dNo, dName)

Course(cNo, cName, dNo) Register(sNo, cNo)



| Student |         |     |
|---------|---------|-----|
| sNo     | sName   | dNo |
| S01     | James   | D01 |
| S02     | Rocky   | D01 |
| S03     | Jackson | D02 |
| S04     | Jane    | D01 |
| S05     | Milli   | D02 |

| Dept |       |
|------|-------|
| dNo  | dName |
| D01  | CSE   |
| D02  | EEE   |

| Course |       |     |
|--------|-------|-----|
| cNo    | cName | dNo |
| C11    | DS    | D01 |
| C12    | OS    | D01 |
| C21    | DE    | D02 |
| C22    | PT    | D02 |
| C23    | CV    | D03 |

| Register |     |
|----------|-----|
| sNo      | cNo |
| S01      | C11 |
| S01      | C12 |
| S02      | C11 |
| S03      | C21 |
| S03      | C22 |
| S03      | C23 |
| S04      | C11 |
| S04      | C12 |
| S05      | C11 |
| S05      | C21 |

SQL Query:

```
SELECT * FROM Student AS S WHERE NOT EXIST
(SELECT cNo FROM Course WHERE dNo = "D01"
EXCEPT
```

```
SELECT cNo FROM Register WHERE sNo = S.sNo)
```

The number of rows returned by the above SQL query is \_\_\_\_\_.

CO3 (b) Schema-1

[08]

Employee(empId, empName, address, Gender, deptNo, Salary)

Department(deptNo, deptName)

Dept\_Locations(deptNo, deptLocation)

Project(pNo, pName, pLocation, deptNo)

Works\_On(empID, pNo, Hours)

Consider the above Schema-1. Write SQL Queries for following:

- Retrieve the names and Ids of employees who work for CE department.
- Retrieve all the employees whose name starts with 'C'.
- Retrieve all the employees whose salary is not between 50,000 and 90,000.
- Retrieve the list of employee names, their department & project names they are working on, ordered alphabetically by department name and within each department, ordered alphabetically by employee name.
- Using IN Operator, find the names and address of the employees who work for the department located in 'Nadiad'.
- Find out names of employees whose salary is greater than all employees in Department No: 3
- Find names of the departments that do not have any employee.
- List the department numbers with average salary greater than 50,000.

- CO3 (c) Consider a database table T containing two columns X and Y each of type integer. [02]  
After the creation of the table, one record (X=1, Y=1) is inserted in the table. Let MX and MY denote the respective maximum values of X and Y among all records in the table at any point in time. Using MX and MY, new records are inserted in the table 28 times with X and Y values being MX+1, 2\*MY+1 respectively. It may be noted that each time after the insertion, values of MX and MY may change. What will be the output of the following SQL query after the steps mentioned above are carried out?

```
SELECT Y FROM T WHERE X=6;
```